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Roll No

BE-3001 (AT/FT)-CBGS

B.E., III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions out of eight.
ii) All questions carry equal marks.

1. State which of the following is true/false.

- i) The coefficients a_0 in Fourier expansion of the function $f(x)$ in interval $(-\pi, \pi)$ is given by

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

- ii) The complex Fourier transform of $f(x)$ is given by

$$\int_{-\infty}^{\infty} e^{isx} f(x) \, dx$$

- iii) $L\{\sinh at\}$ is given by $\frac{s}{s^2 + a^2}$

- iv) For the function $f(z) = \frac{4-3z}{z(z-1)(z-2)}$, $z=2$ is a double pole.

- v) n^{th} approximation through Euler's method is given by

$$y_n = y_{n-1} + h f(x_{n-1}, y_{n-1})$$

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2. a) Obtain Fourier series for $f(x) = x^2$ in $-\pi < x < \pi$.
- b) Develop $\sin\left(\frac{\pi x}{l}\right)$ in a half-range cosine series in the range $0 < x < l$.
3. a) Find the Fourier transform of $f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$
- b) Find the Fourier cosine transform of e^{-x^2} .
4. a) Evaluate the followings:
- i) $L^{-1}\left\{\frac{1}{s^2 + 8s + 16}\right\}$
- ii) $L^{-1}\left\{\frac{s+1}{s^2 + 6s + 25}\right\}$
- b) Evaluate the followings:
- i) $L\{e^{-4t} \sin 3t\}$
- ii) $L\{t \sin 2t\}$
5. a) Using convolution find $L^{-1}\left\{\frac{s^2}{(s^2 + a^2)^2}\right\}$
- b) Using Laplace transform, solve $(D^2 + 1)y = t \cos 2t$, given that $y = Dy = 0$, when $t = 0$

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6. a) Discuss whether $f(z) = \frac{1}{z}$ is analytic or not.
b) Find the order of each pole and residue at if of $\frac{1-2z}{z(z-1)(z-2)}$.
7. a) Show that $u = x^3 - 3xy^2$ is harmonic and find the corresponding conjugate.
b) Using contour integration, evaluate the integral $\int_0^{2\pi} \frac{dv}{2 + \cos \theta}$
8. a) Using Taylor series method, solve $\frac{dy}{dx} = 1 - 2xy$, given that $y(0) = 0$
b) Using Runge-Kutta method of fourth order to solve $\frac{dy}{dx} = xy$ for $x = 1.2$ initially $x = 1, y = 2$ (Take $h = 0.1$).
